

WASP-36b

R-0785

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_241122 · created 2026-07-28T06:41:03+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460636.985 +/- 0.0057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.056 +/- 0.049
Transit depth (Rp/Rs)^2	0.32 +/- 0.55 [%]
Orbital Inclination (inc)	87.99 +/- 2.89
Airmass coefficient 1 (a1)	0.0151 +/- 0.0083
Airmass coefficient 2 (a2)	3.0 +/- 0.71
Scatter in the residuals of the lightcurve fit is	45.16 %
Best Comparison Star	None
Optimal Aperture	2.64
Optimal Annulus	7.0
Transit Duration (day)	0.0557 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.056 ± 0.049 vs 0.1368 (deviation 1.46σ)
Duration fit vs published	0.0557 d vs 0.0758 d (deviation 2.55σ)
Mid-time O–C	-14.3 min
Depth SNR	1.0
Residual scatter	45.16 %

Light curve

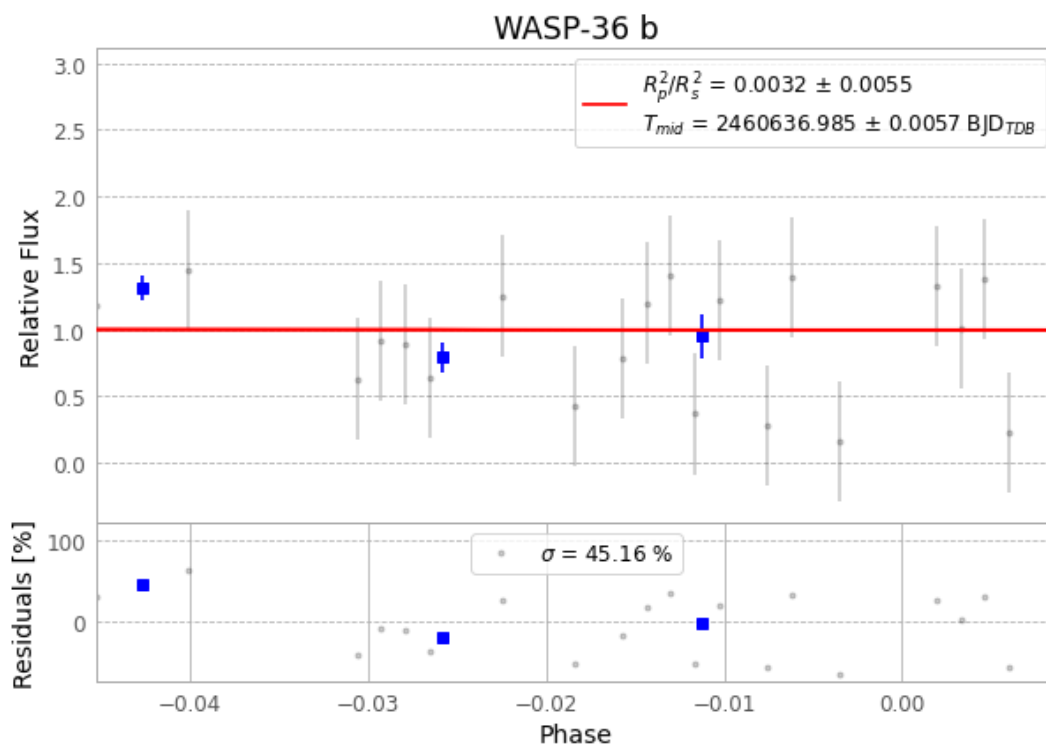


Figure 1 — FinalLightCurve_WASP-36 b_2024-11-22.png (SHA-256 89cc2342414abdf8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [560, 413], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4a83b07e0b630dd8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

41 FITS frames (27 MB), WASP-36241122095447.FITS → WASP-36241122115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-241122.log (7504b86ad2b1...), FinalLightCurve_WASP-36 b_2024-11-22.png (89cc2342414a...), FinalLightCurve_WASP-36 b_2024-11-22.csv (29a489db243f...)

Compute resources

Wall time 90.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 06:41Z.